

K-Theory

An Analytic & Algebraic Mix

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1 Introduction

Welcome! This is a seminar on K-theory and we have two major goals:

1. Cover the fundamentals of algebraic and topological K-theory.
2. To demonstrate that K-theory has applications in unexpected (at least, at first glance) places.

For more details on the exact topics we cover, see 4. For more details on the prerequisites, see 2. K-theory has its origins in the work of Grothendieck on the Riemann-Roch problem and, later, the work of Atiyah & others on the index theory of elliptic operators. However, its reach has expanded significantly since that time, with many other great mathematicians being involved in developing this subject in different directions. We will give a much more detailed introduction to K-theory in the first lecture; see 4.1.

The goal isn't to provide a modern treatment of K-theory; this would require a continuation seminar (and far more background from participants). We are planning for one in Winter 2026/2027, with the goal of reading Alexander Efimov's paper on the K-theory of dualizable presentable ∞ -categories (see [2]). That seminar will, very roughly, cover the following material:

1. The K-theory of Quillen exact categories (roughly, Chapter 3 of [5]).
2. The K-theory of Banach/ C^* -categories.
3. Efimov K-theory, based on the aforementioned paper.

However, there is still time to discuss the contents of the latter seminar. What matters is that this seminar will provide a lot of the intuition for the one we plan to organize next academic year.

2 Organizational Details

The seminar will be held on **Tuesdays**, from 4pm-6pm, in the BMS seminar room at HU Berlin. Since there are going to be participants outside of Germany, we will stream the seminar. **If you are in Berlin or Potsdam, we do recommend coming in person because that makes the seminar far more interactive.** Let us say something about prerequisites. A reasonable amount of mathematical maturity is required. We expect familiarity with the following topics:

1. The basics of commutative algebra.
2. The basics of (general) topology.
3. The basics of analysis.

The basic language for K-theory involves some category theory, so it would be good for you to know this but we will introduce it in supplemental videos that will be made throughout the semester based on request.

We strongly recommend attending this seminar if you have an interest in analysis or analysis-related topics. One of the organizers has an interest in the analytic applications of K-theory and he has made a significant effort to also include such discussions in the seminar. This is also good for students who lean more towards algebraic/geometric topics, since it will expose them to concepts in analysis within a setting which they might find interesting.

For example, we will be discussing Van Daele's K-theory of $\mathbb{Z}_2$ -graded Banach algebras as our main setting for the proof of the Bott Periodicity Theorem. This will, of course, require some familiarity with functional analysis (which we will introduce in supplemental content). Given that we introduce functional analysis for a completely different purpose, we expect that it might be motivating for participants who haven't seen it before and interesting for those who have seen it.

Now, let us say a few words about references. In general, there isn't a single, good book which covers every single thing we wish to discuss. So, we have written down detailed abstracts for the talks in 4 and we have listed specific references that one could/should follow when giving those talks. **If you would like to give a talk and you want to deviate from the suggested plan, then speak to us beforehand.**

Finally, we are fully aware that we might go outside of the bounds of the semester for the seminar. **We’re perfectly happy to do this and if you can’t attend during that time, the lectures that we do hold will be recorded.** Moreover, if the participants would like to see some special topics, then suggestions for these are welcome (for example, specialized talks on number-theoretic applications or C^* -algebraic applications). We only ask that you inform us in advance so we have time to look for speakers/prepare the talks ourselves if we need to do it.

3 Timetable

The following table lists the dates of the lectures, the titles of the lectures and the proposed speakers.

Date	Title	Speaker
14th Oct	An Overview of K-Theory	Abhijeet & Bogdan
21st Oct	The Construction of Algebraic K_0	Abhijeet
28th Oct	The Idempotent Picture of Algebraic K_0	Abhijeet
4th Nov	K_0 of Dedekind Domains	Job Winterdijk
11th Nov	K_0 of Continuous Functions	Jon Righard
18th Nov	The Algebraic Excision Theorem	Abhijeet
25th Nov	The Properties of Topological K^0	Abhijeet
2nd Dec	The Fredholm Index & The Classifying Space of K-Theory	Abhijeet
9th Dec		Bogdan
16th Dec		Bogdan
6th Jan		Bogdan
13th Jan	Relative K_1 & Exact Sequences	Bogdan
20th Jan	Van Daele’s K-Theory I	Abhijeet
27th Jan	Van Daele’s K-Theory II	Abhijeet
3rd Feb	Van Daele’s K-Theory III	Abhijeet
10th Feb	Van Daele’s K-Theory IV	Abhijeet

For detailed descriptions of the expected content of each lecture, see 4.

4 The Fundamentals

In this section, we give an overview of the lectures, providing references and specifics on the material that will be covered in each lecture (the subsections below correspond to one lecture each). We do this to encourage participants to volunteer in giving the lectures since they don’t have to fuss about how to organize the material and they can just focus on presenting the contents well. Here are a few general rules for everyone:

1. The seminar is detail-oriented, so we expect details to be presented.
2. It’s okay to skip **some** specific computations in proofs. However, if you are skipping a detail, then you should explicitly set it as an exercise. You may also give a hint for the exercise so it may be easy for listeners who wish to verify these details for themselves.
3. In general, our seminar will make use of some results which are not directly related to K-theory but which are crucial for making certain computations in K-theory work. It’s acceptable to not present full proofs of such results, though you should be prepared to talk about the mathematical culture surrounding them.
4. We also wish to bring together mathematics students from different parts of math via this seminar. If you have specific applications of K-theory to your area and you feel that one of the talks might allow you to speak about those applications in some detail, then you should ask us if you can do that talk.

With these ideas in mind, let us proceed with describing the contents of the seminar.

4.1 An Overview of K-Theory

We will cover some administrative details for the seminar and begin to give an overview of K-theory as a whole. Abhijeet will start the first half of the lecture by talking about the general structure of K-theory and give an overview of its applications to operator algebras & noncommutative geometry (more specifically, he will talk a bit about the Baum-Connes Conjecture with coefficients). Bogdan will give an overview of its applications to algebraic geometry in the second half of the lecture, starting with a deeper overview of the origins of K-theory in the work of Grothendieck & Serre on the Riemann-Roch problem.

4.2 The Construction of Algebraic K_0

We will cover some basics on projective modules & the Grothendieck construction in the hopes of constructing algebraic K_0 over the category of unital rings. We hope to also compute the K_0 -groups in the two easy cases of fields and PIDs. We will essentially follow [5, p.1-6, p.11].

4.3 The Idempotent Picture of Algebraic K_0

We will present the standard picture of algebraic K_0 before providing another perspective via idempotents. This is extremely useful for computational purposes and we will see that it gives us Morita Invariance & Continuity of algebraic K_0 without much effort. We will follow [5, p. 7-9] and [4, Chapter 1].

4.4 K_0 of Dedekind Domains

We will present a relatively self-contained treatment of K -theory on Dedekind domains. The minimal goal is to set up a relationship between algebraic K_0 and the class group of a Dedekind domain, another invariant whose calculations are quite hard and are of interest in algebraic number theory. We will mainly follow [5, Chapter 1.4, p.16-21], up till the proof of Theorem 1.4.12.

4.5 K_0 of Continuous Functions

We will go through some aspects of the theory of finite-dimensional vector bundles over a compact Hausdorff space, culminating in a proof of the Serre-Swan Theorem. The construction of topological K^0 will also be given; the lecturer should discuss the extension of K -theory to non-unital rings. We will roughly follow [5, p.32-36]. It might be instructive for the lecturer to work through [1, Chapter 8, 119–124] since that does the classical proof of the Bott Periodicity Theorem for topological K -theory.

4.6 The Algebraic Excision Theorem

The three main properties of topological K^0 will be stated:

1. Half-exactness.
2. Excision.
3. Homotopy-Invariance.

To prove these facts, we will need to extend algebraic K_0 to non-unital rings and prove the Algebraic Excision Theorem. We will mainly follow [5, Chapter 1.5, p.27-31]. This marks the end of our exposition on algebraic K_0 for the rest of the seminar.

4.7 The Properties of Topological K^0

The goal is to establish the properties of topological K^0 based on the results of the preceding lecture. For (3), an elementary introduction to the basic theory of $\mathbb{C}$ -Banach algebras will be necessary. So, we require some familiarity with Banach spaces; this will be provided through supplemental videos. We also hope to say a few words about operator K_0 . We will mostly follow [5, p.36-38].

4.8 The Fredholm Index & The Classifying Space of K -Theory

We introduce the definition of a Fredholm operator between $\mathbb{K}$ -Banach spaces, where $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, and prove the Toeplitz Index Theorem, a (very) special case of the Atiyah-Singer Index Theorem. The latter has close connections to K -theory and we provide some references which discuss those connections in detail. Finally, we discuss two important homotopical results:

1. Kuiper's Theorem.
2. The Atiyah-Jänich Theorem.

Both of these results are extremely technical and would likely require several lectures worth of preparation to fully prove. If the audience wishes to see supplemental content regarding them (and, in general, the material in this lecture), then we might consider making such content.

4.9 Construction of Algebraic K_1

We construct the algebraic K_1 functor and establish some of its basic properties in detail, based on Section 1 of [5, Chapter 2.1, p. 59-62]. The main properties are left as exercises and it would be nice to provide sketches during the actual lecture.

4.10 Computations of K_1 -Groups I

We compute the K_1 -groups of division rings and local rings based on [5, Chapter 2.2, p.62-74]. It would be a good idea to cover the proofs up till Corollary 2.2.6 before briefly mentioning local rings arising from operator theory as covered in the exercises. In particular, one should specifically mention if the local ring structure leads to an interesting analytic construction that is useful in operator theory. **This lecture will be done by Bogdan.**

4.11 Computations of K_1 -Groups II

We compute the K_1 -groups of PIDs and Dedekind domains based on [5, Chapter 2.3, p.74-83]. The goal would be to establish the Dirichlet Unit Theorem (Theorem 2.3.8), though it seems that a proof sketch would be more reasonable. It would also be interesting to indicate the representation-theoretic proof (via Kazhdan) of $SK_1(R)$ vanishing when R is the ring of algebraic integers in a number field and use that to compute $K_1(R)$ fully.

4.12 Relative K_1 & Exact Sequences

We define the relative K_1 -groups based on [5, Chapter 2.5, p.92-107]. The main goal is to establish the 6-term exact sequence of algebraic K-theory; this is Theorem 2.5.4, p.95-97. It would also be a good idea to discuss how to find explicit generators for $K_1(R, I)$ if there is time (otherwise, one or two results to this effect should simply be stated).

4.13 Van Daele's K-Theory I

In this lecture, we start working towards a proof of the Bott Periodicity Theorem. We will not follow the original approach as developing K-theory on spaces would require us to deal with analytic technicalities that take away from the main ideas. Instead, we develop Van Daele's K-theory on $\mathbb{Z}_2$ -graded Banach algebras, based on the papers [6] and [7]. In this lecture, we provide some more basic background on Banach algebras while covering sections 1 and 2 of [6].

4.14 Van Daele's K-Theory II

We provide some physical motivation for Van Daele's definition of topological K-theory based on [3] and we cover sections 3 and 4 of [6], ending by establishing the 6-term exact sequence of K-theory and a more explicit description of the boundary homomorphism.

4.15 Van Daele's K-Theory III

We state the Bott Periodicity Theorem, introducing the two main ingredients needed to prove it:

1. Clifford algebras and some basic facts about them (given with proof).
2. The isomorphism from $K(A)$ to $K(S(A) \hat{\otimes} C)$.

The Bott Periodicity Theorem is essentially just a “formal” consequence of these two facts put together; the second fact will be proved in the next lecture. We mainly follow Section 3 of [7].

4.16 Van Daele's K-Theory IV

We prove statement (2) as presented above; this consists of Sections 1 and 2 of [7]. These are actually rather long and the proof is a bit technical, so a summary of the technical bits is expected (with proof sketches if there is time) and the main argument should be presented in detail.

References

- [1] Johannes Ebert. *A Lecture Course on the Atiyah-Singer Index Theorem*. URL: https://ivv5hpp.uni-muenster.de/u/jeber_02/skripten/mainfile.pdf.
- [2] Alexander I. Efimov. *K-theory and localizing invariants of large categories*. 2025. arXiv: 2405.12169 [math.KT]. URL: <https://arxiv.org/abs/2405.12169>.
- [3] Johannes Kellendonk. *On the C^* -algebraic approach to topological phases for insulators*. 2017. arXiv: 1509.06271 [math.KT]. URL: <https://arxiv.org/abs/1509.06271>.
- [4] John Roe. *Lectures on K-Theory and Operator Algebras*. 2017-2018. URL: <https://www.ams.org/open-math-notes/omn-view-listing?listingId=110719>.
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- [6] Alfons Van Daele. “K-theory for graded Banach algebras I”. In: *The Quarterly Journal of Mathematics* 39.2 (June 1988), pp. 185–199. ISSN: 0033-5606. DOI: 10.1093/qmath/39.2.185. eprint: <https://academic.oup.com/qjmath/article-pdf/39/2/185/4396390/39-2-185.pdf>. URL: <https://doi.org/10.1093/qmath/39.2.185>.
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